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Department of Computer Science
Faculty of Engineering, Built Environment & IT
University of Pretoria

COS344 - Computer graphics
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Practical 3 Specification: 3D Rendering

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Start By Date: 24-03-2026

Due Date: 13-04-2026 at 10:00

Total Marks: 100

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1 General Instructions

- *Read the entire assignment thoroughly before you begin coding.*
- This assignment should be completed individually.
- **Every submission will be inspected with the help of dedicated plagiarism detection software.**
- Be ready to upload your assignment well before the deadline. There is a late deadline which is 1 hour after the initial deadline which has a penalty of 20% of your achieved mark. **No extensions will be granted.**
- If your code does not compile, you will be awarded a mark of 0. The output of your program will be primarily considered for marks, although internal structure may also be tested (eg. the presence/absence of certain functions or structure).
- Failure of your program to successfully exit will result in a mark of 0.
- Note that plagiarism is considered a very serious offence. Plagiarism will not be tolerated, and disciplinary action will be taken against offending students. Please refer to the University of Pretoria's plagiarism page at <https://portal.cs.up.ac.za/files/departamental-guide/>.
- Unless otherwise stated, the usage of additional libraries outside of those indicated in the assignment, will **not** be allowed. Some of the appropriate files that you have submit will be overwritten during marking to ensure compliance to these requirements. **Please ensure you use C++11**
- All functions should be implemented in the corresponding `cpp` file. No inline implementation in the header file apart from the provided functions.
- The usage of ChatGPT and other AI-Related software to generate submitted code is strictly forbidden and will be considered as plagiarism.
- The use of this practical, in any form, by any company/business/organisation outside the University of Pretoria is strictly forbidden. This includes, but is not limited to, the use of the practical for discussion sessions, review sessions, marketing tools, or providing certain aspects as a free service.
- No pre-build objects and textures may be used. All objects and textures that you need to use must be created by yourself.
- You must use OpenGL version 3.3 for this practical.

2 Overview

For this practical, you will need to render a low-polygon single-hole miniature golf course with a windmill obstacle in the course. The focus of this practical will be on the windmill and not the course as a whole.

3 Background

Miniature golf, often referred to as mini-golf or putt-putt, originated in the early 20th century as a simplified and recreational adaptation of traditional golf. Designed to be accessible to players of all ages and skill levels, miniature golf courses typically consist of short putting holes featuring artificial turf, obstacles, and themed decorations. These courses emphasise precision and problem-solving rather than power, making them popular in family-oriented and leisure environments.



Figure 1: Example of a miniature golf course from <https://minigolfcreations.com/wp-content/uploads/2018/01/Mini-Golf-Creations-Vines-2.jpeg>

Windmills are mechanical structures that convert wind energy into rotational motion and have historically been used for tasks such as grinding grain, pumping water, and generating power. Their distinctive rotating blades and iconic appearance have made them culturally recognisable symbols, particularly in European landscapes.



Figure 2: Example of a windmill obtained from https://www.designingbuildings.co.uk/w/images/a/ac/Trad_windmill-pixabay.jpg

In miniature golf courses, windmills are commonly used as interactive obstacles rather than functional energy devices. The rotating blades introduce a dynamic challenge by intermittently

blocking the ball's path, requiring players to carefully time their putts. Beyond gameplay, windmills contribute to the visual appeal and thematic storytelling of a course, reinforcing the playful and whimsical atmosphere that characterises miniature golf.



Figure 3: Example of a miniature golf course windmill obtained from https://fairwayminiaturegolf.com/uploads/1/0/4/7/104762931/hole-10_orig.jpg

4 Your Task:

For this practical, you will need to render a low-polygon 3D depiction of a real-world, miniature golf windmill, placed in a basic single-hole course. An example of what you are expected to render is given in Figure 4. In the sections below, the different requirements are laid out. You are allowed

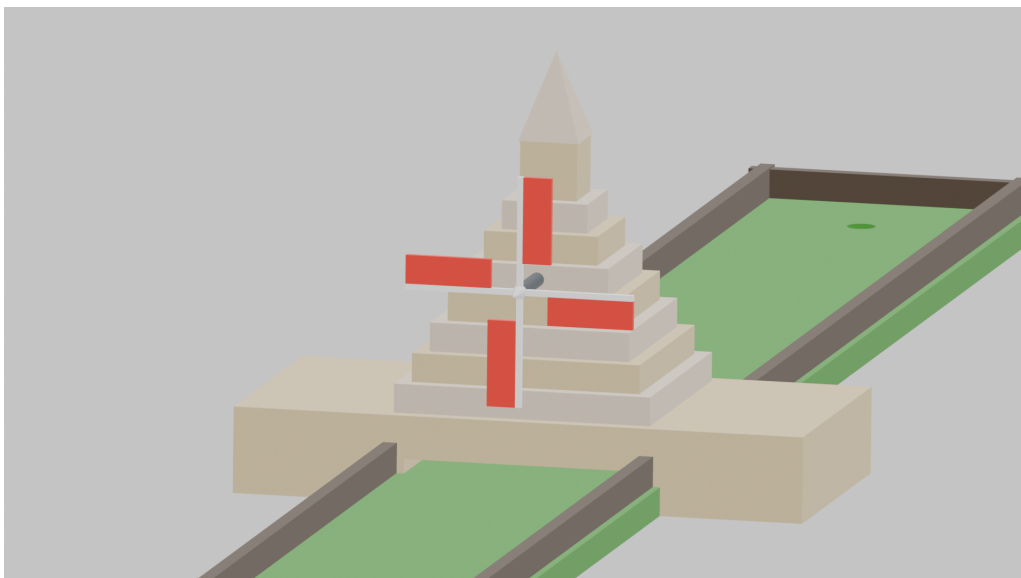


Figure 4: Example of a partially complete expected render

to design any type of windmill as long as it complies with the following requirements. Due to the rendering being a low-polygon depiction of a real-world miniature golf course windmill, you will also need to submit planning photos, which will also be discussed in the sections to come.

4.1 Windmill requirements

The windmill should comply with the following:

- The rendering should look like the windmill you are taking inspiration from, i.e., you cannot render an arbitrary windmill.
- There should be a gap for the ball to pass through.
- The windmill should be attached to the course.
- Should have at least four blades for its rotor.

Your windmill can contain more details; if you wish to add them, just be mindful of the delay in rendering high-polygon shapes.

4.2 Course requirements

The golf course should comply with the following:

- The course should have a boundary wall such that the “ball” cannot go out of bounds.
- The course should have a cutout for where the golf hole will go.
- The course should have a designated starting mat/square.
- The course should have at least two plants/decorations outside of the boundary wall.
- The course should have a grass-like surface where the game of golf can be played.

Your course can contain more details; if you wish to add them, just be mindful of the delay in rendering high-polygon shapes.

4.3 Shape requirements

Your windmill and course are required to at least contain the following distinct¹ 3D shapes:

- Cuboid
- Triangular Prism
- Cone
- Cylinder
 - The two circles at the top and bottom of the cylinder should contain at least 8 vertices.

¹Distinct implies that the shape as a whole is counted and not the internal polygons used to create the shape.

- The cylinders should be used for the axle that connects the rotor to the windmill, and the golf hole, but can be used for other elements as well.

Note, in total, your 3D render should contain at least 20 3D visible shapes using the list above. You can add extra 3D shapes if needed. Also, remember that other shapes will hide some of the faces of the shapes used in the 3D windmill and course. You are also allowed to add decorative 2D shapes (such as a window) on the faces of the 3D shapes to add extra details, but these shapes will not count towards the required 20 3D shapes.

4.4 Colour requirements

The colouring of your windmill and course needs to be inspired by a real-world picture of the windmill you have selected. Your render should consist of at least 6 colours with the following requirements:

- The grass needs to be green.
- The boundary walls need to be a different colour from the rest of the course.
- The blades need to be a distinct colour.
- The starting point mat/section needs to be a distinct colour
- The background colour also has to be distinctive. *Hint: there is an OpenGL function that can assist with this.*
- The axle needs to be a metallic colour.

In conclusion. Your windmill and course need to contain at least 6 different solid, distinctive colours. You are allowed to add extra details to your windmill and course to make it more realistic.

4.5 Transformation requirements

In this section, the transformation or animation requirements for your rendering are discussed. Note keep the animation rates low, such that it allows you to press the key multiple times before going out of camera view.

4.5.1 Rotations

- When the **W** key is pressed, the windmill and course need to rotate about the x -axis.
- When the **S** key is pressed, the windmill and course need to rotate about the x -axis in the opposite direction of the **W** key.
- When the **A** key is pressed, the windmill and course need to rotate about the y -axis.
- When the **D** key is pressed, the windmill and course need to rotate about the y -axis in the opposite direction of the **A** key.
- When the **E** key is pressed, the windmill and course need to rotate about the z -axis.

- When the **Q** key is pressed, the windmill and course need to rotate about the z -axis in the opposite direction of the **E** key.

4.5.2 Translation

- When the **I** key is pressed, both the windmill and the course's y -axis position need to increase.
- When the **K** key is pressed, both the windmill and the course's y -axis position need to decrease.
- When the **L** key is pressed, both the windmill and the course's x -axis position need to increase.
- When the **J** key is pressed, both the windmill and the course's x -axis position need to decrease.
- When the **O** key is pressed, both the windmill and the course's z -axis position need to increase.
- When the **U** key is pressed, both the windmill and the course's z -axis position need to decrease.

4.5.3 Rotor and blades

As the rotor and blades are a vital part of a windmill, your windmill's rotor and blades will also need to turn to simulate the rotor rotating. This implies that you will need to rotate the rotor and blades around an arbitrary axis. Rotor and blade rotation must appear realistic. To add extra complexity, the rotor and blades' speed should be controlled by the user as described below:

- Initially, the rotor and blades' speed should be 0.
- When the **+** key is pressed, the rotor and blades' speed should increase.
- When the **-** key is pressed, the rotor and blades' speed should decrease.
- *Hint: Ensure the propeller does not rotate backwards with multiple - key presses.*

Thus, the more the **+** key is pressed, the faster the rotor and blades should spin and vice versa for the **-**.

4.5.4 Examples

Below are examples of the result after multiple key presses, assuming the aircraft is at the original rendering position.

Note that when a key is pressed, the windmill and course should **not** return the original position. Your program should be able to apply the transformation in any arbitrary order with the windmill, course, rotor and propeller at any position on the screen.

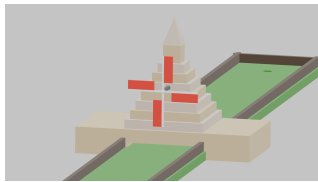


Figure 5: Original

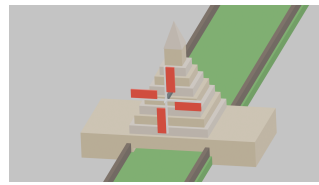
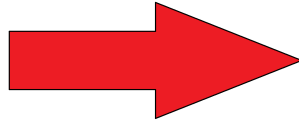


Figure 6: After a series of W key presses

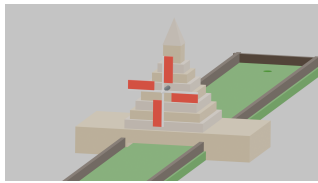


Figure 7: Original

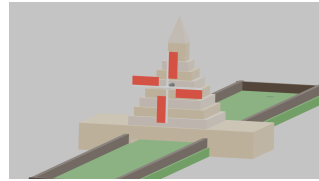
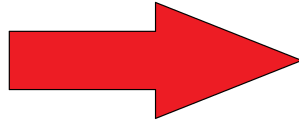


Figure 8: After a series of S key presses

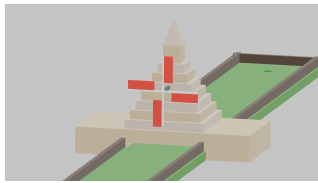


Figure 9: Original

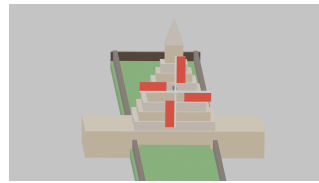
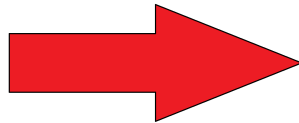


Figure 10: After a series of A key presses

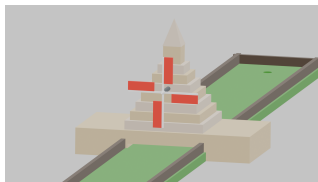


Figure 11: Original

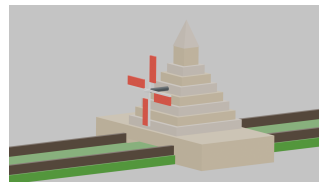
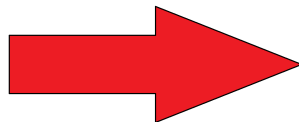


Figure 12: After a series of D key presses

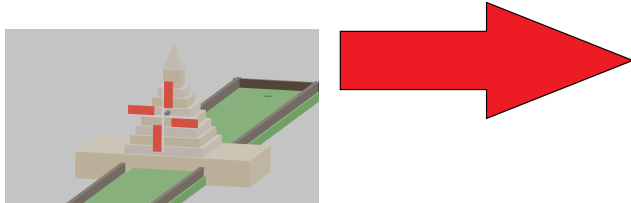


Figure 13: Original

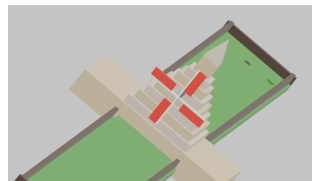


Figure 14: After a series of E key presses

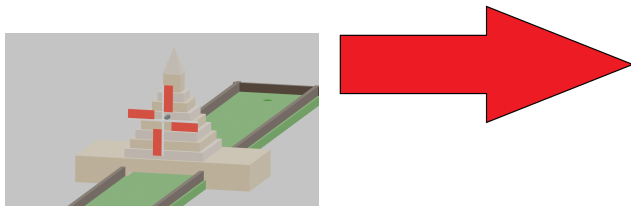


Figure 15: Original



Figure 16: After a series of Q key presses

4.6 Wireframe

You are also required to implement a wireframe for your windmill and course. The wireframe should maintain the colour scheme of your windmill and course, such that the colours of the different shapes can be identified. The wireframe should also conform to all the transformations described earlier.

Your program should toggle between the wireframe and the normal windmill and course when the **Enter** key is pressed. *Hint: you may need to implement a time delay between key presses for the wireframe toggling, so that the expected behaviour is achieved.*

Please note that you **must** use the `GL_LINES` to implement your wireframe. **Using the `glPolygonMode` function will result in the forfeiting of your wireframe marks.**

4.7 Reference and planning

Due to the requirement that the windmill is based on a real-world miniature golf course windmill, you will need to provide some planning or a reference photo that you will show during the marking of the practical. You need to include the following photos with your submission:

Below you will find the reference photo that was used to create the example windmill and course.



Figure 17: Augmented photo showing the planning of the 3D low polygon rendering

The planning photos will contribute **6 marks** to your final mark for this practical. If no photo is included, you will receive 0 for this section.

5 Marking rubric

The following rubric will be used to mark your submitted assignment. Note you will be demoing the practical during the practical sessions on your own computer or a lab computer. Please see Table 1 for the rubric. Note: 1 mark will be subtracted for each transformation if the render is moved back to the centre before a new transformation is applied. Although not included in the marking rubric, as stated in Section 4.6, the inclusion of the planning photos will contribute 5 marks to the final mark of the practical.

6 Bonus marks

There are 10 bonus marks available. Bonus marks are worth 2 for each extra thing that you render or implement. Some examples of things that count as extra are:

- Adding details on the windmill, such as doors, windows or text.
- Creating a rotor windmill.

- Creating a windmill that contains an internal tunnel instead of a straight gap for the ball to pass through.
- Additional details to the course, like sand pits, flowers, or rivers.
- Adding a 3D spherical golf ball to the render.
- Making the course have an incline and a decline.

7 Implementation Details

- You need to use OpenGL version 3.3 for this practical.
- You may **not** use any of the build-in mathematical libraries within the `glm` package. This included matrix arithmetic. *Hint: You may use your practical 1 in this practical.*
- You may **not** use any of the built-in OpenGL functions to generate the shapes for you. You need to create each shape from first principles.
- You may **not** use any of the built-in OpenGL functions to perform the transformations of the shapes. You need to transform each shape from first principles, either explicitly or by using the matrix arithmetic techniques discussed in class.
- You may only use the following C++ and OpenGL libraries:
 - `stdio.h`
 - `stdlib.h`
 - `iostream`
 - `iomanip`
 - `initialisation list`
 - `cmath`
 - `sstream`
 - `GL/glew.h`
 - `GLFW/glfw3.h`
 - `glm/glm.hpp`

You may also use the `shader.hpp` and `glad.c` files that assist with compiling and linking of shaders. Your code should be able to be compiled without the assistance of an IDE, i.e. the project needs to be able to be compiled from terminal.

- All your helper classes and files needs to be in the same directory of the `main.cpp`.
- Ensure that the title of the window of your program is your correct student number.

8 Submission

You are required to submit on ClickUp under the appropriate submission link. In the archive that you submit, include a makefile and compiling instructions such that the program can be compiled and executed by the markers if needed. *Failure to upload to ClickUP will result in you forfeiting all marks for this practical.* No exceptions will be made on this matter.

9 Demo Instructions

1. You will first be required to download your submission from ClickUP.
2. You will then demo your practical to the tutor.
3. In the presence of the tutor, you will be required to upload the archive you downloaded from ClickUP to FitchFork. *Failure to upload to FitchFork will result in you forfeiting all marks for this practical.* No exceptions will be made on this matter.

Assessment Criteria	0	1	2	3	4
Windmill requirements [10 marks]					
Realistic windmill	No resemblance of actual windmill		Slight resemblance to windmill		Realistic resemblance to windmill
Golf ball gap	There is no gap		There is a gap		
Windmill attached to course	Not attached		Windmill is attached		
Four bladed rotor	There is no rotor	There is a rotor but not enough blades	There is a rotor with sufficient blades		
Course requirements [14 marks]					
Boundary wall	No boundary wall		Partial boundary wall		Complete boundary wall
Golf ball gap	There is no gap		There is a gap		
Designated starting mat	No mat		Mat is present		
At least two plants/decor	There are no plants/decor		There is a single plant/decor		There are sufficient plants/decor
Grass like course surface	No course surface		Course surface is present		
Shape requirements [12 marks]					
At least one cubiod	There are no cuboids	There is at least one cuboid			
At least one triangular prism	There are no triangular prisms	There is at least one triangular prism			
At least three cylinders	There are no cylinders	There are cylinders but of the wrong vertex count	At most one cylinder that has the correct number of vertices		There are at least two cylinders that has the correct number of vertices
At least one cone	There are no cones	There are cones but they do not have the correct vertex count	There is at least one cone that has the correct vertex number		
At least 20 distinct 3D shapes	There are no 3D shapes	There are at most 5 3D shapes	There are at most 10 3D shapes	There are at most 15 3D shapes	There are at least 20 3D shapes
Colour requirements [14 marks]					
Green grass	Not correct colour		Correct colour		
Distinct boundary wall colour	Non-distinct colour		Correct colour		
Distinct blade colour	Non-distinct colour		Correct colour		
Distinct starting matt colour	Non-distinct colour		Correct colour		
Metallic rotor axel colour	Non-distinct colour		Correct colour		
Background colour is distinctive	Not distinctive colour or black		Correct colour		
At least 6 colours used	Only a single colour used			Correct number of colours	
Rotation requirements[12 marks]					
X-axis rotation	The render is not able to rotate around the x-axis		The render is able to only rotate in one direction around the x-axis		The render is able to rotate in both directions around the x-axis
Y-axis rotation	The render is not able to rotate around the y-axis		The render is able to only rotate in one direction around the y-axis		The render is able to rotate in both directions around the y-axis
Z-axis rotation	The render is not able to rotate around the z-axis		The render is able to only rotate in one direction around the z-axis		The render is able to rotate in both directions around the z-axis
Transformation requirements[12 marks]					
Horizontal translations	The render is not able to move along the x-axis		The render is able to only to move along the x-axis in one direction		The render is able to move along the x-axis in both directions
Depth translations	The render is not able to move along the z-axis		The render is able to only to move along the z-axis in one direction		The render is able to move along the z-axis in both directions
Vertical translations	The render is not able to move along the y-axis		The render is able to only to move along the y-axis in one direction		The render is able to move along the y-axis in both directions
Rotor and blades rotation[8 marks]					
Rotor and blades de-acceleration	No rotor or blade is turning		The rotor and blade rotates at a constant speed		The rotor and blade rotates at a increasable speed
Rotor and blades acceleration	No rotor or blade is turning		The rotor and blade rotates at a constant speed		The rotor and blade rotates at a decreasesable speed
Wireframe[12 marks]					
Render	No wireframe		Partial wireframe		Correct wireframe
Colour	No wireframe		Partially coloured		Correctly coloured
Transformations	No wireframe	Partial transformations			Correct transformations

Table 1: Marking rubric